

Question 2 — Non-Blocking Multi-Pattern Timing & Debouncing

Register configuration:

PB0–PB3 are configured as outputs for the LEDs. PD2 is used as the push button input, with its internal pull-up resistor enabled. Because of the pull-up resistor, the button reads LOW when pressed. The LED states are controlled directly through PORTB using bit masking instead of `digitalWrite()`.

Timing calculations:

LED A stays ON for 200 ms and OFF for 800 ms, so its full cycle is 1000 ms. LED B changes state every 350 ms, giving a 700 ms full cycle. LED C starts ON for 150 ms and then stays OFF for 650 ms, giving an 800 ms cycle. Button changes are accepted only if the input remains stable for at least 40 ms to remove bouncing. If the button is held for more than 2000 ms, LED D starts toggling every 50 ms until the button is released. The system status is printed to the Serial Monitor every 2500 ms.

System behavior:

Using separate `millis()` timers allows all LED patterns and button operations to run at the same time without blocking the program. A normal button press toggles LED D once. If the button is held for more than 2 seconds, LED D enters the fast flashing mode until release.